Chemical Reaction Engineering

Professor: Meenesh R Singh

- What is Chemical Reaction Engineering?
    1. Chemical Reaction + Engineering

- What is chemical reaction?
    1. A chemical reaction is a process of chemical changes that involves the breaking and/or forming of chemical bonds to create new substances (compounds).
        - A chemical bond forms/breaks when atoms transfer or share electrons.
        - Example:
            - A covalent bond is formed when atoms share electrons to complete outermost shell.
            - Atoms can gain and lose electrons to complete outermost shell and become ions. Oppositely charged ions forms ionic bonds. (Ionic bonds do not form compounds)
            - Metal atoms can donate all d-electrons to electron cloud creating attractive forces between electron cloud and metal ion. This is metallic bond
            - Coordinate bond is a covalent bond in which both electron came from the same atom.
        - Examples of the chemical reaction
            - $NaHCO_3 + CH_3COOH \rightarrow H_20 + CO_2 + CH_3COONa$

- Important features of Chemical Reactions
    1. Number of atoms of each element in reactants and products are same.
        - $NaHCO_3 + CH_3COOH \rightarrow H_20 + CO_2 + CH_3COONa$

          3 C atoms = 3 C atoms

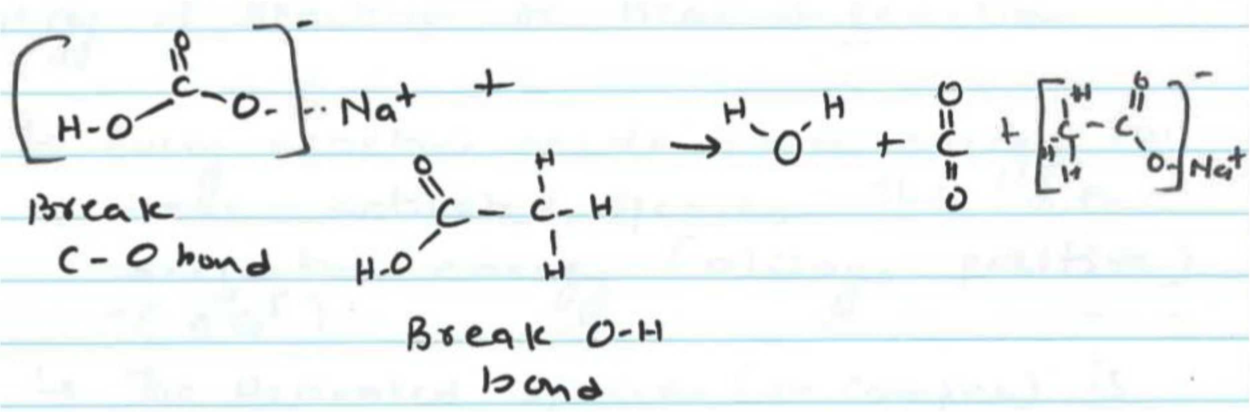
          5 O atoms = 5O atoms

          5 H atoms = 5H atoms

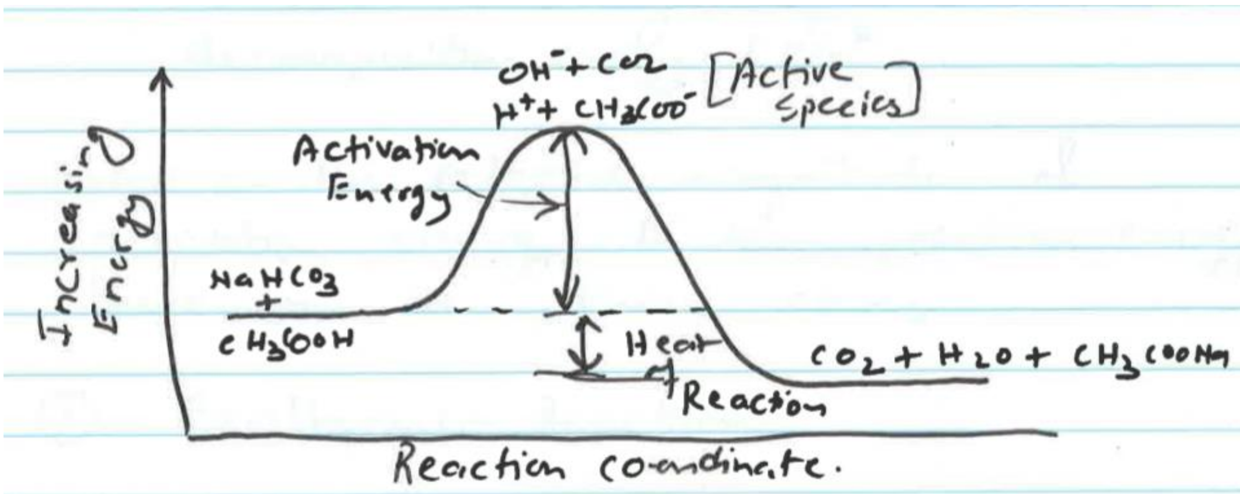
          1 Na atoms = 1 Na atom
    2. Since chemical reactions only involves electron transfer, the mass of reactants is equal to the mass of products. Change in mass is only associated with the reaction involving change in nuclei. Ex: nuclear reaction.

- Balancing Chemical Reaction
    1. Write each product and reactant using its chemical formula.
       Ex: reactants: NH_3, CO_2. Product: $(NH_2)_2CO$
       $NH_3 + CO_2 \rightarrow (NH_2)_2CO$
    2. Add integer coefficient to the reactants and products to balance number of atoms.
       $2NH_3 + 1CO_2 \rightarrow 1(NH_2)_2CO + 1\ H_2O$

- Stoichiometry
  - The coefficients used for balancing the chemical reaction is called the stoichiometric coefficient.
    - 1 molecule of urea is produced for every 2 molecules of NH_3 and 1 molecule of CO_2 consumed.
    - We can also say that 1 mole of urea is pro produced for every 2 moleof NH_3 and 1 mole of CO_2 consumed. [1 mole = 6.022×10^{23} molecules]
- Types of Chemical Reactions
  1. Addition or Synthesis Reaction (Electrophilic and Nucleophlic)
     Ex: $4Fe + 3O_2 \rightarrow 2\,Fe_2O_3$
     $nC_2H_4 \rightarrow [-CH_2\text{-}CH_2-]_n$
  2. Decomposition Reactions
     Ex: $2NaHCO_3 \rightarrow CO_2 + Na_2CO_3 + H_2O$ (under heat)
  3. Single displacement reactions
     Ex: $Fe_{(s)} + CuCl_{2(aq)} \rightarrow Cu_{(s)} + FeCl_{2(aq)}$
  4. Double- displacement reaction
     Ex: $CaCO_3$ + H2SO4 $\rightarrow CaSO_4 + H_2O + CO_2$ [precipitation reaction]
     $HCl + NaOH \rightarrow NaCl + H_2O$ [acid-base reaction]
  5. Combustion reaction
     Ex: $2\,C_8H_{18} + 25\,O_2 \rightarrow 16\,CO_2 + 18\,H_2O$
  6. Oxidation reaction (redox) reaction
     Ex: $H_2 + F_2 \rightarrow 2HF$ [H is oxidized, its oxidation number increase from 1 to +1. F is reduced, its oxidation number decreases from 0 to -1]
- Will there be a chemical reaction? When and why a reaction occurs? Let's revisit the reaction.



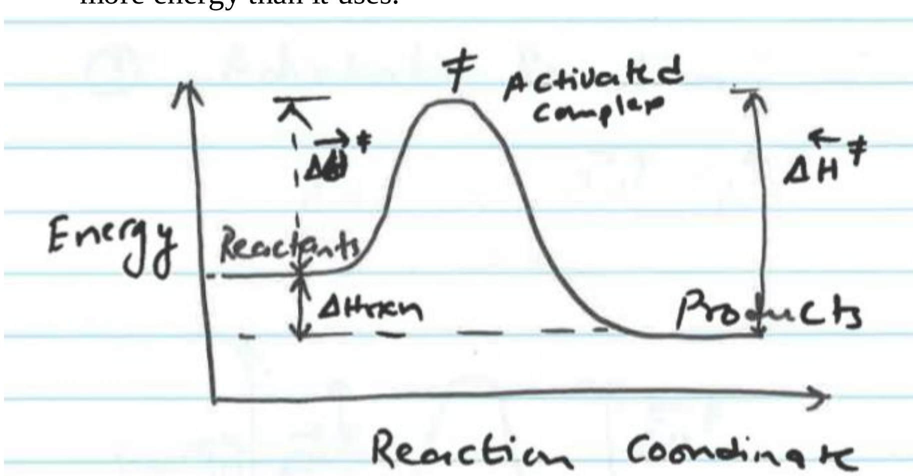
- To initiate this reaction some energy must be supplied to break C-O and O-H bond.



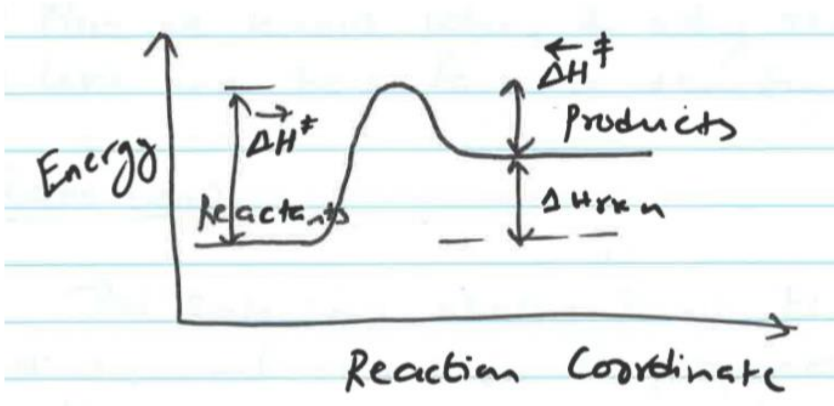
Also, NaHCO3 and CH_3COOH must be in close proximity.

- If the energy supplied is higher than the activation energy, the reaction will be favorable.
- The reaction will eventually occur to minimize the energy of the system.

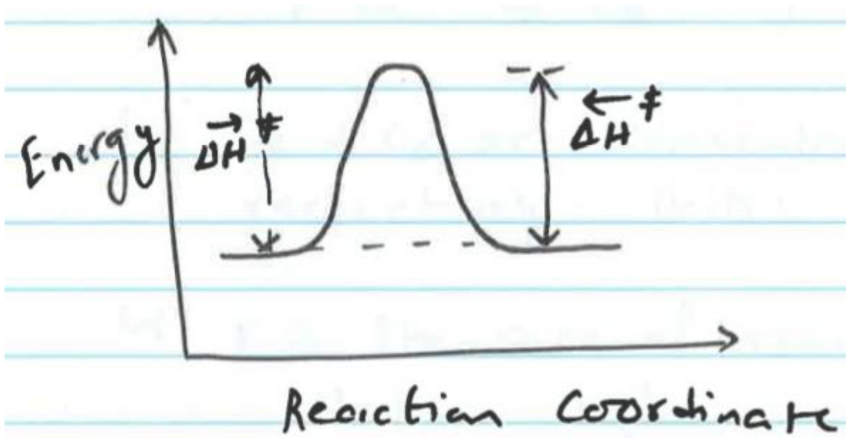
- Energy of Reaction or Heat of Reaction
  - Every reaction needs some energy to make activated species. This is the activation energy (always positive) $(\Delta \boldsymbol{G}^{\neq})$
    The activated species (or complex) is not stable (it contains higher energy). It releases energy to become more stable products. Since the energy is released, it is negative. This is the decomposition energy. $(\Delta \boldsymbol{G}^{\neq})$
  - Based on the relative magnitude of activation energy and decomposition energy there can be there cases.
    - Exothermic Reaction: Heat of reaction: $\Delta H_{rxn} = H_{products} - H_{reactants} < 0$. Heat is released. → A reaction is exothermic if it is releases more energy than it uses.



- Endothermic Reaction: $\Delta H_{rxn} = H_{products} - H_{reactants} > 0$. Heat is absorbed. → A reaction is endothermic if it is uses more energy than it releases.



- Adiabatic Reaction: $\Delta H_{rxn} = 0$. A reaction is adiabatic if it neither absorbs nor releases energy.

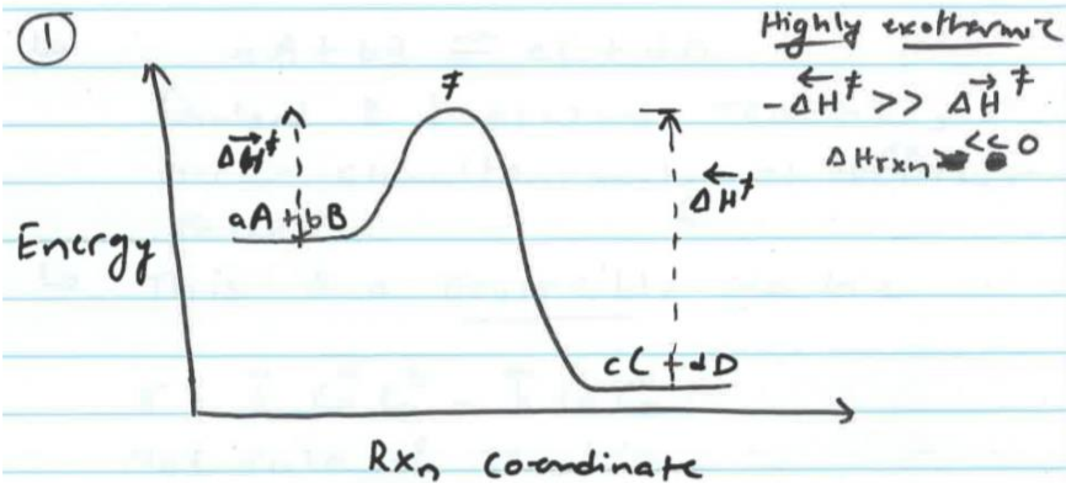


Now we know when and why reaction occurs. Let's see how fast a reaction occurs.

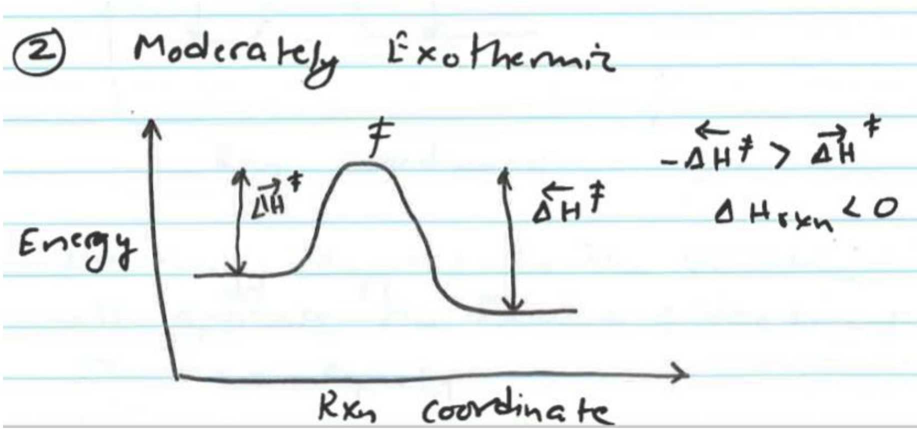
- Rate Law: The rate law states that the rate of a chemical reaction is proportional to the product of concentration of reactants with each concentration raised to a power equal to the stoichiometric coefficient of the reaction.
  - Ex: aA + bB → cC + dD
    Rate of reaction $r \propto C_A^a C_B^b$.
    C_A^a, C_B^b are concentration of A and B respectively. Unit mole/m^3.
    r is rate of reaction. Unit mole/(m^3.s)
    The proportionality constant also referred to as a rate constant is K. unit: 1/s.
    → $r = K\ C_A^a C_B^b$
    a and b are raction order with respect to A and B respectively.

K can be a function of temperature, pressure, concentration electric potential ect.

- Irreversible vs reversible reaction
  - Consider highly exothermic vs moderately exothermic reactions.
  - Exothermic reaction



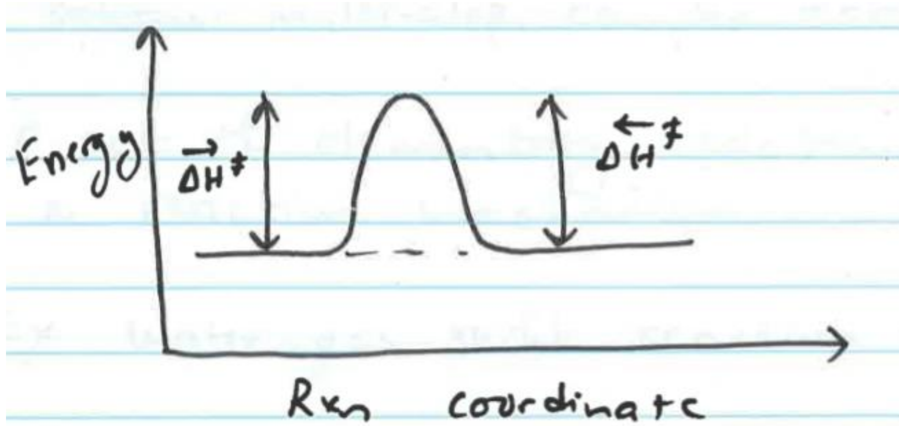
    - Since $-\Delta H^{\neq}_{backward} \gg \Delta H^{\neq}_{foward}$. It is almost impossible to convert C and D to A and B using the energy required to convert A and B to C and D.
    - Such reactions are referred to as an irreversible reactions.
  - Moderately Exothermic



    - The forward barrier $\Delta H^{\neq}_{foward}$ is comparable to $-\Delta H^{\neq}_{backward}$, such that the energy supplied for aA + bB → cC + dD
    - If aA + bB ↔ cC + dD. Forward and backward reactions occur simultaneously at different rate. This is a reversible reaction. Net rate:

$$r = K_{forward} C_A^a C_B^b - K_{backward} C_C^c C_D^d$$

- Chemical Equilibrium
  - Consider a case when $-\Delta H^{\neq}_{backward} = \Delta H^{\neq}_{foward}$



The energy supplied to the reaction will activate the forward and backward reaction equally.

Such that, $r = 0 = K_{forward}C_A^a C_B^b - K_{backward}C_C^c C_D^d$

$$\frac{K_{forward}}{K_{backward}} = \frac{C_C^c C_D^d}{C_A^a C_B^b} = K\ (constant\ )$$

This is the law of mass action. K is the equilibrium constant given as K = exp$\left[-\frac{\Delta G_{rxn}}{RT}\right]$

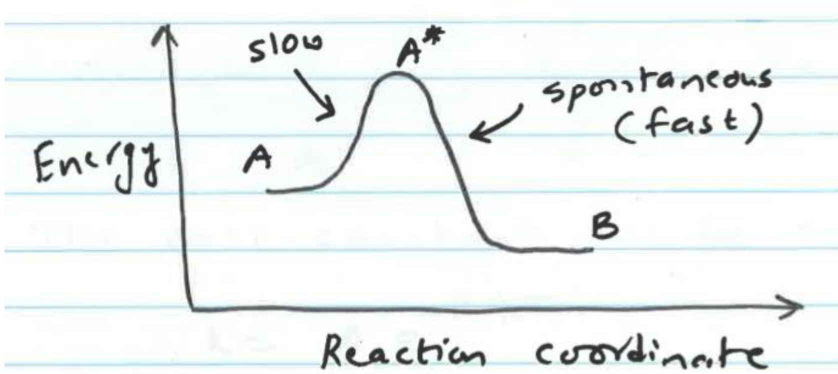
- Elementary Reaction
  - Elementary reaction is the most fundamental single step reaction in a given multi-step, complex reaction. A set of elementary reaction forms a reaction mechanism. For example: Water-gas shift reaction. $CO + H_2O = CO_2 + H_2$
    - This is a multi-step, complex reaction. The elementary reactions are usually postulated based on an step-by-step bond breaking and forming process.

CO + * → *CO	H_2O + * → *H_2O
*CO + *OH → *COOH + *	*H_2O → *OH + *H
*COOH → *CO_2 + *H	*H + *H → *H_2
*CO_2 → * + CO_2	*H_2 → * + H_2

There are 4 elementary reactions in water-gas reaction.

  - Types of elementary reaction
    - Unimolecular reaction. Ex: *COOH → *CO_2 + *H
    - Bimolecular reaction. Ex: *CO + *OH → *COOH + *
    - Termolecular reaction. Ex: *CO + *OH + *H →*CO_2 + H_2

- Rate of elementary reaction
  - Consider an elementary reaction A→ B.



The elementary reaction A→ B can be written as A = A* → B
The first reaction is slower than the second reaction. But both reactions occurs at the same rate. Therefore, the first reaction can be considered at equilibrium such that

$$\frac{C_A^*}{C_A} = K = exp\left[-\frac{\Delta G_f^{\neq}}{RT}\right] \rightarrow \; C_A^* = C_A xp\left[-\frac{\Delta G_f^{\neq}}{RT}\right]$$

The rate of second reaction is

$$r = \; K_0 C_A^* = \; K_0 C_A exp\left[-\frac{\Delta G_f^{\neq}}{RT}\right]$$

$$r = \; K_0 e^{+\Delta S_f^{\neq}/R} \; e^{-\Delta H_f^{\neq}/RT} C_A$$

The rate constant can be expressed as k = A$e^{-E/RT}$. This is Arrhenius Law.
Prefactor A can be estimated using collision theory and transition state theory.
Reaction barrier $\Delta H_f^{\neq}$ can be estimated using semi-empirical approach or quantum mechanical methods.